
Formatting Instructions For NeurIPS 2025

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Abstract

1

2 1 Introduction

3 In the high stakes world of algorithmic decision making, the outcomes of machine learning models
4 can have significant impacts on individuals and society in areas such as healthcare, criminal justice,
5 lending, hiring. In this contexts, fairness, is not just a desirable property but a critical requirement
6 TODO: [ADD CITATIONS]. In particular, group fairness, which requires predictive behaviour
7 remains equitable across demographically defined groups, has emerged as a key focus of research in
8 this area.

9 Notwithstanding the significant progress in developing group fairness algorithms, most of the existing
10 work assumes a stationary data distribution which, in the real world, rarely holds. Instead, data
11 distributions evolve over time: policy shifts, seasonal effects and changes in societal norms can lead
12 to distribution shifts (often referred to as concept drift) that can undermine predictive performance
13 and fairness guarantees.

14 While online learning algorithms have been created to preserve predictive accuracy, there has been
15 less work on how to preserve fairness under concept drift. This is a critical gap, because fairness
16 violations may remain undetected longer than accuracy degradation, disrupting the trustworthiness of
17 machine learning systems and leading to significant harms.

18 To address this challenge it is important not only to detect distributional changes as they occur but, at
19 the same time, characterize their impact on fairness and react accordingly. In this paper, we study
20 fairness-aware learning under concept drift in online settings and investigate how a novel framework
21 for detecting and responding to concept drift can be used to preserve fairness guarantees over time.

22 2 Experiments

Table 1: Experimental results on the Adult dataset across four drift scenarios. Aranyani maintains superior accuracy while effectively mitigating fairness violations (DP) during drift events.

Model	No Drift (Acc $\uparrow$)	Abrupt (Acc $\uparrow$)	Gradual (Acc $\uparrow$)	Reversal (Acc $\uparrow$)
ARF	0.8350 ± 0.0000	0.8100 ± 0.0000	0.8400 ± 0.0000	0.8200 ± 0.0000
RFR	0.8038 ± 0.0074	0.8025 ± 0.0043	0.8063 ± 0.0054	0.7888 ± 0.0096
Aranyani	0.8538 ± 0.0035	0.8607 ± 0.0172	0.8573 ± 0.0022	0.8516 ± 0.0045

Model	No Drift (DP $\downarrow$)	Abrupt (DP $\downarrow$)	Gradual (DP $\downarrow$)	Reversal (DP $\downarrow$)
ARF	0.1040 ± 0.0000	0.1154 ± 0.0000	0.0890 ± 0.0000	0.1154 ± 0.0000
RFR	0.0592 ± 0.0071	0.0639 ± 0.0017	0.0668 ± 0.0106	0.0753 ± 0.0111
Aranyani	0.0860 ± 0.0022	0.0414 ± 0.0167	0.0129 ± 0.0063	0.0854 ± 0.0056

Table 2: Folktables dataset results (mean $\pm$ std over seeds)

Model	Accuracy $\uparrow$	DP $\downarrow$
Aranyani		
ARF	0.735 ± 0.0000	0.0750 ± 0.0000
RFR	0.747 ± 0.0175	0.0257 ± 0.0168

23 **References**

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26 is permissible to reduce the font size to `small` (9 point) when listing the references. Note that the
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28 [1] Alexander, J.A. & Mozer, M.C. (1995) Template-based algorithms for connectionist rule extraction. In
29 G. Tesauro, D.S. Touretzky and T.K. Leen (eds.), *Advances in Neural Information Processing Systems 7*, pp.
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32 *GENeral NEural Simulation System*. New York: TELOS/Springer-Verlag.

33 [3] Hasselmo, M.E., Schnell, E. & Barkai, E. (1995) Dynamics of learning and recall at excitatory recurrent
34 synapses and cholinergic modulation in rat hippocampal region CA3. *Journal of Neuroscience* **15**(7):5249-5262.

35 **A Technical Appendices and Supplementary Material**

36 Technical appendices with additional results, figures, graphs and proofs may be submitted with
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38 ZIP file below before the supplementary material deadline. There is no page limit for the technical
39 appendices.

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72 Answer: [TODO]

73 Justification: [TODO]

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